

THREAT INTELLIGENCE SERIES

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SANDWORM

SUMMARY

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DOSSIER 1907

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01 EXECUTIVE SUMMARY

Sandworm is a cyber unit of Russia's GRU military intelligence directorate, publicly identified by the U.S. Department of Justice as Unit 74455. Unlike most state-sponsored groups, which prioritize covert, long-term espionage, Sandworm specializes in disruption—targeting physical infrastructure, economies, and public confidence.

In December 2015 and again in 2016, the group cut electricity to hundreds of thousands of Ukrainian civilians using malware built to manipulate power-grid control systems. In June 2017, it escalated further: a wiper disguised as ransomware, delivered through a compromised software update, escaped its intended target and became the costliest cyberattack ever recorded.

WARNING

NotPetya was never meant to stay contained. It crippled hospitals, shipping companies, manufacturers, and logistics firms across the globe—most of which were never the intended target.

02 KEY FINDINGS

- Caused two confirmed power outages in Ukraine using custom grid-sabotage malware.
- Launched NotPetya through a supply-chain compromise of Ukrainian tax software.
- Embedded false-flag code in the 2018 Olympic Destroyer attack to misdirect investigators toward North Korea and China.
- Six GRU officers were publicly indicted by the U.S. Department of Justice, with a \$10 million reward still offered for information leading to their capture.

03 TECHNICAL ANALYSIS

Sandworm's earliest known activity dates to around 2009, when it used early BlackEnergy malware against government and energy targets in Ukraine and Eastern Europe. Its defining moment came in December 2015, when BlackEnergy and the KillDisk wiper were used to cut power across parts of Ukraine—the first confirmed cyberattack to cause a blackout. A year later, the group returned with Industroyer, malware designed specifically to communicate with industrial power-grid protocols.

Its most destructive operation came in June 2017. After compromising the update server of the Ukrainian accounting software M.E.Doc, Sandworm distributed NotPetya to thousands of organizations. Although it appeared to be ransomware, the malware permanently destroyed data regardless of payment. Using the EternalBlue exploit, it spread rapidly beyond Ukraine into multinational networks.

The U.S. Department of Justice documented severe disruption at Heritage Valley Health System in Pennsylvania, where cardiology, radiology, nuclear medicine, and surgical systems were unavailable for about a week, while administrative systems remained affected for nearly a month. Global losses exceeded \$10 billion, making NotPetya the most expensive cyberattack on record.

Beyond Ukraine, Sandworm conducted the 2017 hack-and-leak operation targeting Emmanuel Macron's presidential campaign. In 2018, it launched Olympic Destroyer against the PyeongChang Winter Olympics, embedding code intended to imitate Chinese and North Korean threat actors in an effort to delay attribution.

04 ASSESSMENT

DOSSIER 1907 assesses that Sandworm represents a distinct doctrine in state-sponsored cyber operations: disruption is the objective, not the by-product. While many APT groups seek persistence and secrecy, Sandworm's operations are often designed to be visible, destructive, and psychologically impactful.

Although six officers were indicted in 2020, none have been apprehended. The unit has remained active since Russia's 2022 invasion of Ukraine, continuing to target critical infrastructure. NotPetya demonstrated that Sandworm accepts global collateral damage as an operational risk rather than a constraint—a reality organizations should consider when operating near, or supplying technology into, active conflict zones.

05 REFERENCES

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3. MITRE ATT&CK® Enterprise Framework, Sandworm Team (G0034).